

5.2 Organization of this International Standard

This International Standard groups software system life cycle activities into seven process groups, each with a defined purpose, desired outcomes, and related activities. These process groups collectively form a Process Reference Model (PRM) that can be applied to any system or software life cycle model, providing a structured yet adaptable framework.

The seven process groups are:

- a) Agreement Processes (subclauses 5.2.2.1.1 and 6.1)
- b) Organizational Project-Enabling Processes (subclauses 5.2.2.1.2 and 6.2)
- c) Project Processes (subclauses 5.2.2.1.3 and 6.3)
- d) Technical Processes (subclauses 5.2.2.1.4 and 6.4)
- e) Software Implementation Processes (subclauses 5.2.2.2.1 and 7.1)
- f) Software Support Processes (subclauses 5.2.2.2.2 and 7.2)
- g) Software Reuse Processes (subclauses 5.2.2.2.3 and 7.3)

These process groups are introduced below and depicted in Figure 1.

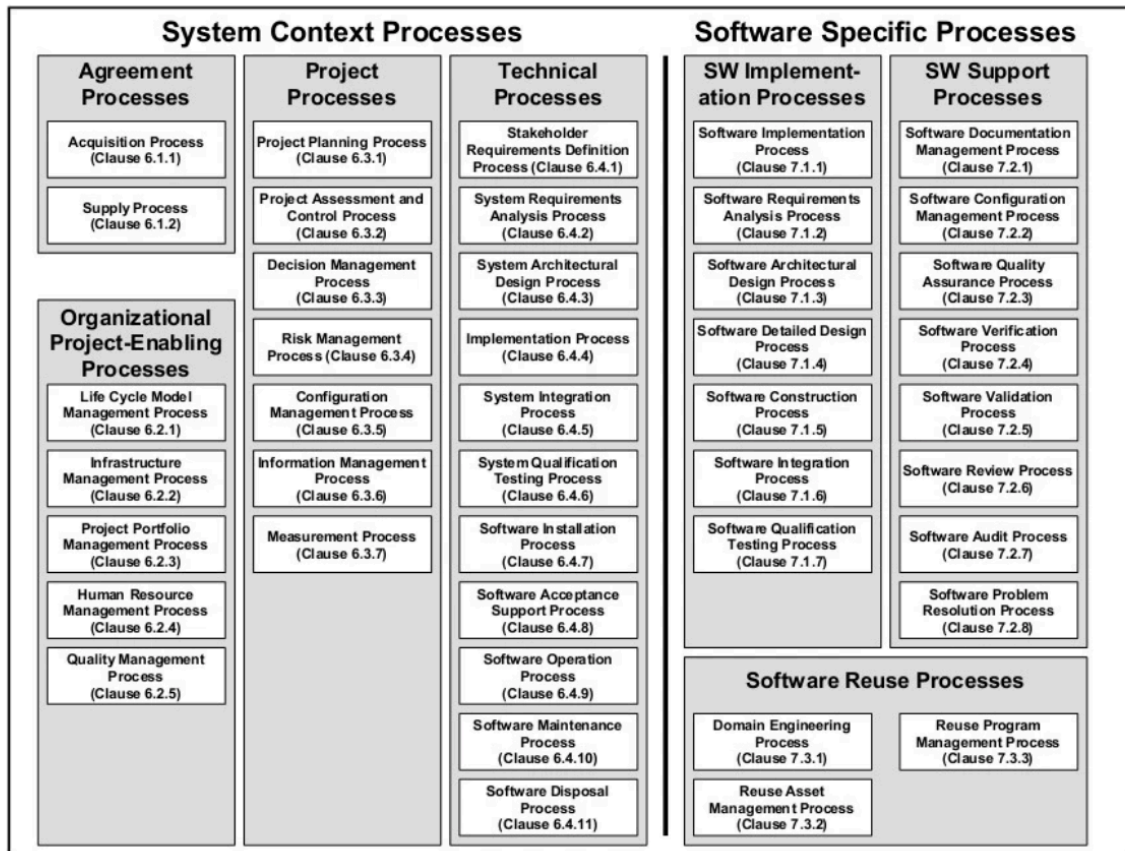


Figure 1 — Life Cycle Process groups

Figure 1 — Relationship among Process Groups

Figure 1 illustrates how these process groups interact and support each other throughout the life cycle. It shows the organizational processes enabling projects, project processes managing execution, and technical and software processes realizing and supporting system elements.

Clauses in this International Standard follow a structured numbering system:

- **6.a** and **7.a** denote a process group.
- **6.a.b** and **7.a.b** denote a specific process within that group.
- **6.a.b.1** and **7.a.b.1** describe the process's purpose.
- **6.a.b.2** and **7.a.b.2** describe the process's outcomes.
- **6.a.b.3.c** and **7.a.b.3.c** list process activities.
- **6.a.b.3.c.d** and **7.a.b.3.c.d** list tasks for each activity.

This structure supports clarity, traceability, and alignment between system and software processes. The Process Reference Model does not prescribe a specific system/software life cycle model or implementation method; it is **tailored** based on business needs, application domain, project scale, and risk. Projects typically adopt the organization's defined process in the context of customer requirements.

Process outcomes serve as objective evidence for assessing process capability and identifying improvement opportunities.

5.2.2 Summary of Life Cycle Processes

This International Standard has two major subdivisions:

- **Clause 6** provides a system context for standalone software products, services, or systems.
- **Clause 7** specifies software-specific processes for implementing software elements within larger systems.

To support concurrent application of ISO/IEC 15288 and ISO/IEC 12207, corresponding processes in Clause 6 share the same subclause number (6.x.x). The processes in this International Standard are **software-appropriate specializations or contributions** to the outcomes of ISO/IEC 15288 processes. While many are structurally similar, important distinctions remain regarding purpose, outcomes, and intended audience.

Tailoring: Organizations may tailor processes according to project characteristics, regulatory environment, and business needs, as described in Annex A.

5.2.2.1 System Context Processes

5.2.2.1.1 Agreement Processes define the activities to establish an agreement between two organizations. The Acquisition Process enables conducting business with suppliers for an operational system. The Supply Process facilitates projects resulting in products or services delivered to the acquirer. These Agreement Processes are software-appropriate specializations of those in ISO/IEC 15288.

5.2.2.1.2 Organizational Project-Enabling Processes manage the organization's capability to acquire and supply products or services through project initiation, support, and control. They provide resources and infrastructure to support projects and ensure organizational objectives are met. These processes are not a comprehensive set of business processes.

The Organizational Project-Enabling Processes include:

- Life Cycle Model Management Process
- Infrastructure Management Process
- Project Portfolio Management Process
- Human Resource Management Process
- Quality Management Process

These processes are software-appropriate specializations of the corresponding processes in ISO/IEC 15288.

5.2.2.1.3 Project Processes

This International Standard describes processes for planning, assessing, and controlling projects. These principles apply to any area of an organization's management.

There are two categories of project processes:

1. Project Management Processes: These are used to plan, execute, assess, and control project progress.

- Project Planning Process: Establishes and evolves project plans.
- Project Assessment and Control Process: Assesses actual progress.

2. Project Support Processes: These provide focused tasks for specialized management objectives. They are evident in any undertaking, from organizations to single life cycle processes.

- Decision Management Process: Manages decisions.
- Risk Management Process: Manages risks.
- Configuration Management Process: Manages configurations.
- Information Management Process: Manages information.
- Measurement Process: Measures progress.

The Project Support Processes in this International Standard are similar to those in ISO/IEC 15288, except for formatting differences. Software Supporting Processes may relate to Project Supporting Processes.

5.2.2.1.4 Technical Processes

Technical Processes define system requirements and transform them into products, enabling consistent realization, use, support, and retirement. They ensure that products and services meet requirements for timeliness, cost-effectiveness, functionality, reliability, safety, security, environmental compatibility, and societal expectations.

The Technical Process group includes:

- Stakeholder Requirements Definition
- System Requirements Analysis
- System Architectural Design
- System Implementation
- System Integration

- System Qualification Testing
- System Installation
- System Transition
- System Operation
- System Maintenance
- System Disposal

Software processes such as Software Requirements Analysis, Software Architectural Design, Software Detailed Design, Software Construction, Software Integration, Software Qualification Testing, Software Installation, Software Operation, Software Maintenance, and Software Disposal are **specializations or contributors** to these technical processes.

5.2.2.2 Software-Specific Processes

5.2.2.2.1 Software Implementation Processes

Software Implementation Processes produce a specified software item by transforming defined behavior, interfaces, and constraints into an implemented element that satisfies system requirements. This group includes:

- Software Requirements Analysis
- Software Architectural Design
- Software Detailed Design
- Software Construction
- Software Integration
- Software Qualification Testing

These processes correspond to the Implementation Process in ISO/IEC 15288, specialized for software.

5.2.2.2.2 Software Support Processes

Software Support Processes provide focused activities that ensure software quality and conformity. These processes support the implementation and maintenance of software across its life cycle:

- Software Documentation Management
- Software Configuration Management
- Software Quality Assurance
- Software Verification
- Software Validation
- Software Review
- Software Audit

- Software Problem Resolution

5.2.2.2.3 Software Reuse Processes

Software Reuse Processes focus on systematic reuse of software components and knowledge to improve efficiency, reduce cost, and enhance quality. Operating primarily outside individual projects, this process group includes:

- a) Domain Engineering Process
- b) Reuse Asset Management Process
- c) Reuse Program Management Process

5.2.3 Process Reference Model

Annex B defines the Process Reference Model (PRM) at a higher abstraction level than the detailed requirements in this International Standard. It specifies the **purpose and outcomes** of each process, enabling process capability assessment.

Organizations may evaluate their processes against the PRM, focusing on achievement of purposes and outcomes, without necessarily conforming to the detailed activity descriptions.

NOTE 1 — The term “Process Reference Model” has the same meaning as in ISO/IEC.

NOTE 2 — ISO/IEC 15504-2 defines requirements for assessing processes using the PRM.

For **mapping between system (Clause 6) and software (Clause 7) processes**, see Annex C.